

Patient Shortage in Histopathology: The Cohort’s Between-Patient Spectrum Is Not the Bottleneck of Cohort Whitening

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Abstract

Three cohort-only remedies for patient shortage — whitening along the cohort’s between-patient eigenbasis, directional shrinkage of the class centres, and training against class-centre uncertainty — all build on the same estimate, the between-patient covariance of five patients per class, and all stop at about 2.3 points or below on TCGA-UT. That covariance is a rank-120 estimate in 2,560 dimensions, and sample spectra at such ratios are known to be over-dispersed. This report asks whether the cohort’s *eigendirections* are useful while its *eigenvalues* are too noisy to whiten with. The cohort basis is kept and its eigenvalues λ_j are replaced by λ_j^β for $\beta \in \{0, 0.25, 0.5, 0.75\}$, with $\beta = 1$ being cohort whitening itself, and a validation-tuned arm selects β per fit. Two further arms separate the mechanism: an oracle arm keeps the cohort basis and sets each eigenvalue to the training pool’s between-patient variance along that direction, and a control replaces the basis by the top within-patient principal components of the same rank. On TCGA-UT the tuned arm gains -0.00 points over cohort whitening at five patients (95% interval: -0.01 to 0.00) and fails the prespecified one-point criterion; validation keeps $\beta = 1$ in 29 of 30 fits. Compression is monotonically harmful: $\beta = 0$ costs 0.92 points (0.72 to 1.11). The oracle spectrum gains 0.08 points (0.03 to 0.13), which is 2.8 percent of the 2.66-point gap between cohort-basis and pool-basis whitening, while the missing directions account for 2.58 points (2.25 to 2.89). The empirical exponent relating pool variance to cohort eigenvalue is 0.82, so the cohort spectrum is over-dispersed as expected, but correcting it buys almost nothing. The rank-matched within-patient control matches cohort whitening to within 0.17 points on either dataset, so the specific between-patient directions are not required either. On BRACS no contrast is distinguishable from zero. Second-moment remedies estimated from the cohort alone are therefore limited by which directions the cohort can see, not by how well it scales them.

1 Motivation

Cohort-only remedies for patient shortage have converged on the same ceiling. Whitening the features along the eigen-directions of the cohort’s own pooled between-patient covariance gains 2.26 percentage points at five patients per class on TCGA-UT (Queisler, 2026e); directional shrinkage of the class centres along the same directions gains 1.29 points (Queisler, 2026d); training against the class-centre uncertainty built from the same covariance gains 2.34 points (Queisler, 2026c). An isotropic perturbation of matched total size gains nothing (Queisler, 2026c), so the direction structure of that covariance carries the effect. All three remedies stay far below the 10.66 points of pool-centre headroom (Queisler, 2026b) and far below whitening along the training pool’s between-patient basis, which gains 4.92 points at the same patient count but needs all training patients and is therefore a mechanism reference rather than a remedy.

The estimate these remedies share is extreme in shape. Pooling the deviations of patient means from their class mean over C classes at $C(G - 1)$ degrees of freedom gives a covariance of

rank at most 120 on TCGA-UT and 28 on BRACS at $G = 5$ patients, inside a 2,560-dimensional feature space. Sample covariance spectra in this regime are not merely noisy but systematically spread: the sample eigenvalues of a population with equal variances follow the Marčenko–Pastur law rather than concentrating at the population value (Marchenko and Pastur, 1967), and the largest ones are biased upward (Johnstone, 2001). Covariance estimation for high-dimensional problems responds by shrinking the spectrum while keeping the eigenvectors: linear shrinkage pulls all eigenvalues toward their mean (Ledoit and Wolf, 2004), and nonlinear shrinkage moves each eigenvalue by an amount that depends on its position in the spectrum (Ledoit and Wolf, 2020). Both corrections leave the eigenvectors of the sample covariance untouched, which is exactly the split this report needs.

The question is therefore: *are the cohort’s patient directions useful while its eigenvalues are too dispersed to whiten with, so that compressing the spectrum recovers further headroom?* Cohort whitening hosts the test, because it is the strongest cohort-only remedy and, unlike the centre-uncertainty loss, carries no open grid boundary. The answer is judged at five patients per class, separately on each dataset, against the one-point practical threshold used throughout this series. Two further arms make a negative answer interpretable: an oracle arm that replaces the cohort’s eigenvalues by the quantity they estimate, and a control that replaces the cohort’s directions while keeping everything else.

2 Spectrum model

Write the patch features of class c as $\mathbf{x}_{ik} = \boldsymbol{\mu}_c + \mathbf{u}_i + \mathbf{e}_{ik}$, with class centre $\boldsymbol{\mu}_c$, patient offset $\mathbf{u}_i$, and within-patient residual $\mathbf{e}_{ik}$ (Queisler, 2026b). A cohort of G patients per class estimates the between-patient covariance of the offsets by pooling the deviations of its patient means from their class means over all C classes,

$$\hat{\boldsymbol{\Sigma}}_B = \frac{1}{C(G-1)} \sum_{c=1}^C \sum_{i=1}^G (\mathbf{m}_{ci} - \hat{\boldsymbol{\mu}}_c)(\mathbf{m}_{ci} - \hat{\boldsymbol{\mu}}_c)^\top = \mathbf{U}^\top \text{diag}(\boldsymbol{\lambda}) \mathbf{U}, \quad (1)$$

where the rows $\mathbf{u}_j$ of $\mathbf{U}$ are the eigenvectors with non-negligible eigenvalues $\lambda_1 \geq \dots \geq \lambda_r > 0$ and $r \leq C(G-1)$. Cohort whitening transforms every feature vector by

$$\mathbf{x} \mapsto \mathbf{x}(\mathbf{I} + \kappa \mathbf{U}^\top \text{diag}(\boldsymbol{\nu}) \mathbf{U})^{-1/2}, \quad \kappa = \frac{\text{factor}}{\bar{\nu}}, \quad (2)$$

which multiplies the component along $\mathbf{u}_j$ by $(1 + \kappa \nu_j)^{-1/2}$ and leaves the orthogonal complement unchanged. The spectrum $\boldsymbol{\nu}$ is the object under study. Cohort whitening uses $\boldsymbol{\nu} = \boldsymbol{\lambda}$; pool-basis whitening uses the pool’s eigenbasis and eigenvalues in place of $\mathbf{U}$ and $\boldsymbol{\lambda}$ (Queisler, 2026e). Dividing by the mean $\bar{\nu}$ makes κ scale-free, so any spectrum is comparable to any other at the same grid factor, and a change that rescales all eigenvalues by a constant has no effect.

Three modifications of $\boldsymbol{\nu}$ are tested, all with $\mathbf{U}$ fixed. The *power family* sets $\nu_j = \lambda_j^\beta$. With $\beta = 1$ it is cohort whitening; with $\beta = 0$ it makes the shrinkage factor equal on all r directions, so the transform becomes a uniform contraction of the cohort’s patient subspace and keeps only the subspace, not its internal structure; intermediate β interpolate, in the same spirit as the shrinkage estimators cited above, though as a one-parameter family rather than an optimal rule. The *oracle spectrum* sets

$$\nu_j = v_j = \mathbf{u}_j^\top \boldsymbol{\Sigma}_{\text{pool}} \mathbf{u}_j, \quad (3)$$

the training pool’s between-patient variance along the cohort’s own direction $\mathbf{u}_j$, which is the population quantity λ_j estimates. This arm is a mechanism reference, not a remedy, because it reads the pool. It bounds what a correction of the spectrum alone can achieve within this family, and together with pool-basis whitening it splits the gap between cohort-basis and pool-basis whitening into a spectrum part and a direction part. The relation between the two spectra

is summarized by the *oracle exponent*, the ordinary least-squares slope of $\log v_j$ on $\log \lambda_j$ across the r directions of one fit: a slope below one means the cohort spectrum is more dispersed than the pool variances it estimates, which is the condition that motivates $\beta < 1$.

The final modification changes the directions instead. The *within-patient control* replaces $\mathbf{U}$ by the top r eigenvectors $\mathbf{V}$ of the cohort’s pooled within-patient patch covariance, computed from patch deviations around each patient’s own mean, and keeps the cohort eigenvalues $\boldsymbol{\lambda}$ in rank order. It is a cohort-only construction of the same rank that whitens high-variance directions of the same data without using the between-patient structure. A random subspace of rank 120 in 2,560 dimensions would overlap the cohort basis only negligibly and would reproduce the untransformed fit, so it would not test anything; the within-patient control instead asks whether the specific between-patient directions matter or whether any high-variance directions of matched rank would do. Its agreement with the cohort basis is measured by the subspace overlap $\|\mathbf{U}\mathbf{V}^\top\|_F^2/r$, which is one for identical subspaces and about $r/2,560$ for unrelated ones.

3 Design

The study keeps the setting of Queisler (2026b) on TCGA-UT and of Queisler (2026a) on BRACS. TCGA-UT contributes 30 cancer types and BRACS its seven lesion classes, both with frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), three fixed patient-disjoint splits, and multinomial logistic regression whose regularization strength is selected on validation patient-macro balanced accuracy. For each split and each of ten draws, twenty eligible patients per class are sampled, and nested cohorts of five patients with 32 patches, ten with sixteen, and twenty with eight are formed, so every classifier trains on 160 patches per class at a single patient count for all classes. No additional patients, labels, features, or foundation models enter, and the reused arms are the exact fits of the source studies on the same cohorts, so every contrast is paired draw by draw.

Six new arm families enter at each $G \in \{5, 10, 20\}$ (Table 1). Each of them whitens by Equation 2 with its own spectrum or basis and then fits an ordinary multinomial logistic classifier on the transformed features, so the arms differ from cohort whitening in the whitening spectrum alone, except for the control, which differs in the basis. Every new arm tunes the scale factor $\kappa \in \{0.1, 1, 10, 100\}/\bar{\nu}$ jointly with the regularization strength on validation patient-macro balanced accuracy, the grid and the tie rules of Queisler (2026e). The tuned family Bt fits nothing of its own: for each fit it reads the validation score of the κ that each power arm selected, adds the score of the reused cohort-whitening fit as the $\beta = 1$ candidate, and copies the winning record wholesale, with ties resolved toward the larger β so that a departure from $\beta = 1$ requires a strict validation improvement. The β grid was fixed before fitting and was not extended after test results were seen; whether values above one would be worth testing is addressed by the oracle exponent rather than by a test-set search.

The eigen-decompositions are computed once per cohort and patient count and are shared by the arms that use them, and the whitening is applied through the low-rank basis, so no $2,560 \times 2,560$ matrix is formed. Per dataset, the study fits six families at four scale factors for each of three patient counts and 30 split-draw pairs, that is 2,160 whitened fits, each tuning nine regularization values; the tuned family adds no fit. Selections were written to disk before any test prediction, and test accuracy never entered selection.

4 Evaluation

The endpoint is patient-macro balanced accuracy on the test partition, pooled over splits and draws with equal weight per split. Uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split over 2,000 replicates; all arms share the replicates, so every contrast is paired, and intervals are 95 percent percentile intervals.

Arm	Basis	Spectrum	Role
<i>RG</i>	—	—	Ordinary fit on the real cohort (reused)
<i>CG</i>	—	—	Cohort with pool class centres (reused headroom reference)
<i>RWcG</i>	cohort U	λ_j	Cohort whitening, the host remedy (reused)
<i>RWG</i>	pool	pool	Pool-basis whitening (reused mechanism reference)
<i>FG</i>	cohort U	λ_j^0	Flat spectrum: subspace only
<i>BaG</i> , <i>BbG</i> , <i>BcG</i>	cohort U	$\lambda_j^{0.25}, \lambda_j^{0.5}, \lambda_j^{0.75}$	Power compression
<i>BtG</i>	cohort U	λ_j^β	β selected on validation from $\{0, 0.25, 0.5, 0.75, 1\}$
<i>OG</i>	cohort U	v_j	Oracle spectrum, Equation 3 (mechanism)
<i>PG</i>	within-patient V	λ_j	Rank-matched direction control

Table 1: All new families whiten by Equation 2 and differ from cohort whitening only in the spectrum, except P, which differs only in the basis. *RG* and *CG* bracket the pool-centre headroom; *RWcG* and *RWG* bracket what whitening achieves from the cohort and from all training patients. *OG* and *RWG* read the training pool and are not remedies.

The primary contrast is *Bt5*–*RWc5*, read separately on each dataset. Spectrum regularization counts as a practical success on a dataset if the estimated gain over cohort whitening is at least one percentage point and its paired interval lies entirely above zero; a claim across both cohorts requires success on both. Secondary quantities are the direction-only comparison *F5*–*RWc5*, the spectrum headroom *O5*–*RWc5*, the missing-direction headroom *RW5*–*O5*, the subspace specificity *RWc5*–*P5*, the β profile at every patient count, and the recovery $(\text{BtG} - \text{RG})/(\text{CG} - \text{RG})$ of the pool-centre headroom at five and ten patients. Four readings were fixed in advance. A large *Bt*–*RWc* would mean that the cohort locates its patient subspace but not the strengths along it, which is the case nonlinear shrinkage is designed for. *Bt*≈*RWc* together with *O*≈*RWc* would mean that the spectrum is not the bottleneck and that moment-based cohort-only remedies are exhausted. *O*≫*RWc* with *Bt*≈*RWc* would mean that the spectrum matters but that power compression is the wrong family. *RWc*≈*P* would mean that the between-patient directions are not specific, and that any high-variance subspace of the same rank whitens as well.

5 Results

Spectrum regularization fails the prespecified criterion on both datasets. On TCGA-UT the tuned arm gains -0.00 points over cohort whitening at five patients per class (95% interval: -0.01 to 0.00), and on BRACS -0.11 points (-0.51 to 0.20); neither reaches the one-point threshold, and neither interval lies above zero (Table 2). The tuned arm is not merely flat but nearly identical to its host, because validation selects the uncompressed spectrum $\beta = 1$ in 29 of 30 TCGA-UT fits at five patients, in which case the arm copies the cohort-whitening record. Compression in the direction the over-dispersion argument suggests is harmful throughout on TCGA-UT: -0.16 points at $\beta = 0.75$, -0.42 at $\beta = 0.5$, -0.77 at $\beta = 0.25$, and -0.92 at $\beta = 0$, every interval below zero. On BRACS every power arm lies within 0.35 points of cohort whitening with intervals that include zero, which is unsurprising there, since cohort whitening itself gains only 0.32 points (-0.36 to 1.22) over the real cohort on that dataset.

Arm family	$G = 5$	$G = 10$	$G = 20$
<i>TCGA-UT</i>			
Real cohort (R)	62.48 [61.24, 63.65]	67.79 [66.47, 69.00]	71.89 [70.55, 73.06]
Flat spectrum, $\beta = 0$ (F)	63.82 [62.56, 64.98]	68.32 [67.01, 69.52]	72.04 [70.71, 73.23]
$\beta = 0.25$ (Ba)	63.97 [62.73, 65.12]	68.62 [67.31, 69.83]	72.41 [71.06, 73.59]
$\beta = 0.5$ (Bb)	64.32 [63.09, 65.48]	69.07 [67.79, 70.24]	72.87 [71.55, 74.06]
$\beta = 0.75$ (Bc)	64.58 [63.34, 65.75]	69.47 [68.17, 70.65]	73.09 [71.77, 74.27]
Cohort whitening, $\beta = 1$ (RWc)	64.74 [63.52, 65.90]	69.64 [68.34, 70.84]	73.16 [71.86, 74.34]
Tuned β (Bt)	64.74 [63.51, 65.90]	69.64 [68.33, 70.83]	73.14 [71.83, 74.31]
Oracle spectrum (O)	64.82 [63.59, 65.99]	69.66 [68.35, 70.86]	73.18 [71.88, 74.36]
Within-patient control (P)	64.91 [63.64, 66.09]	69.41 [68.06, 70.62]	73.11 [71.72, 74.33]
Pool-basis whitening (RW)	67.40 [66.14, 68.52]	71.66 [70.39, 72.84]	74.44 [73.10, 75.59]
Pool centres (C)	73.14 [71.62, 74.50]	74.47 [73.01, 75.78]	75.55 [74.11, 76.85]
Gain Bt – RWc	−0.00 [−0.01, 0.00]	−0.01 [−0.02, 0.00]	−0.02 [−0.05, 0.00]
Gain Bt – R	2.26 [1.97, 2.53]	1.85 [1.54, 2.13]	1.25 [1.02, 1.47]
<i>BRACS</i>			
Real cohort (R)	36.18 [33.46, 38.80]	38.49 [35.87, 41.22]	40.97 [38.50, 43.62]
Flat spectrum, $\beta = 0$ (F)	36.28 [33.64, 38.85]	38.45 [35.89, 41.07]	41.06 [38.53, 43.73]
$\beta = 0.25$ (Ba)	36.41 [33.83, 39.03]	38.60 [35.99, 41.27]	40.94 [38.43, 43.58]
$\beta = 0.5$ (Bb)	36.49 [33.89, 39.13]	38.60 [36.03, 41.27]	41.11 [38.56, 43.74]
$\beta = 0.75$ (Bc)	36.42 [33.86, 39.08]	38.66 [35.97, 41.33]	41.24 [38.70, 43.77]
Cohort whitening, $\beta = 1$ (RWc)	36.49 [33.87, 39.24]	38.61 [35.94, 41.34]	41.28 [38.70, 43.88]
Tuned β (Bt)	36.38 [33.76, 39.01]	38.67 [36.03, 41.32]	41.13 [38.66, 43.73]
Oracle spectrum (O)	36.55 [33.97, 39.12]	38.85 [36.21, 41.53]	41.25 [38.73, 43.81]
Within-patient control (P)	36.60 [33.92, 39.21]	38.72 [36.04, 41.45]	40.69 [38.16, 43.40]
Pool-basis whitening (RW)	39.50 [36.82, 41.98]	40.13 [37.39, 42.77]	41.25 [38.79, 43.76]
Pool centres (C)	44.63 [41.33, 48.05]	45.11 [41.85, 48.51]	45.50 [42.20, 48.76]
Gain Bt – RWc	−0.11 [−0.51, 0.20]	0.06 [−0.27, 0.38]	−0.15 [−0.59, 0.25]
Gain Bt – R	0.21 [−0.39, 0.94]	0.18 [−0.39, 0.77]	0.16 [−0.46, 0.87]

Table 2: Compressing the cohort spectrum never helps and, on TCGA-UT, costs up to 0.92 points, while the oracle spectrum on the same directions adds at most 0.08. Accuracy entries are test patient-macro balanced accuracy in percent; gains are paired contrasts in percentage points; bold entries are the primary contrast. Intervals are 95% bootstrap intervals over test patients and draws. The R, C, RWc, and RW rows are the fits of Queisler (2026b), Queisler (2026e), and Queisler (2026a).

Because the tuned arm copies cohort whitening in almost every fit, it inherits its headroom recovery exactly: 0.212 (0.183 to 0.241) of the pool-centre headroom at five TCGA-UT patients and 0.277 (0.231 to 0.325) at ten, against 0.212 and 0.277 for cohort whitening itself, and a share of 0.107 (0.076 to 0.137) of the five-to-twenty patient gap. On BRACS the recovery is 0.024 (−0.054 to 0.101) at five patients and is indistinguishable from zero at every patient count.

5.1 The shape of the β profile

The TCGA-UT profile is monotone in β at all three patient counts and reaches its maximum at the boundary $\beta = 1$ (Figure 1). The cost of compression grows smoothly as the spectrum is flattened, and it is largest at ten patients, where the flat arm loses 1.33 points (1.07 to 1.60) against 0.92 at five and 1.12 at twenty. Half of the whitening benefit survives complete flattening: the flat arm still gains 1.34 points over the real cohort at five patients, that is 59 percent of the 2.26 points of cohort whitening, so most of what whitening does is captured by contracting the cohort’s patient subspace uniformly, and the internal structure of that subspace contributes the rest.

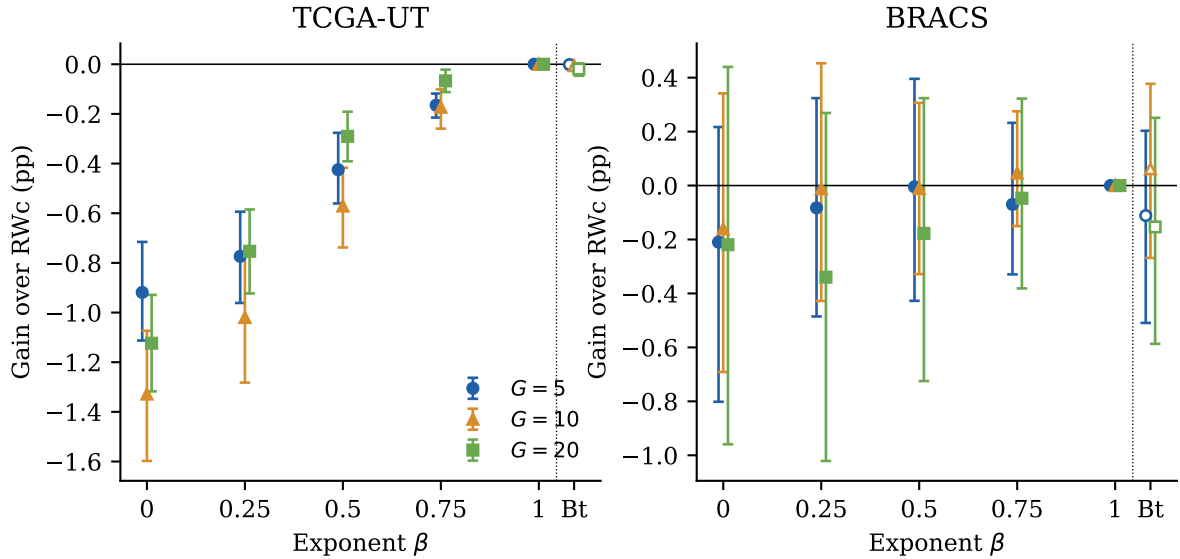


Figure 1: Compressing the cohort’s between-patient spectrum only costs accuracy; the validation-tuned arm returns to the uncompressed spectrum. Paired gain over cohort whitening in percentage points against the exponent β , with $\beta = 1$ being cohort whitening itself and therefore zero by construction. The open markers right of the dotted line are the validation-tuned arm Bt. Error bars show the paired 95% bootstrap intervals; markers are offset horizontally to separate the patient counts.

The BRACS profile is flat within its intervals, which span about one point at $\beta = 0$ and narrow toward $\beta = 1$. Nothing in it separates the exponents, and the widest departure from cohort whitening, −0.34 points at $\beta = 0.25$ and twenty patients, has an interval from −1.02 to 0.27.

5.2 Spectrum against directions

The oracle arm bounds what the spectrum can contribute, and the bound is small. On TCGA-UT it gains 0.08 points over cohort whitening at five patients (0.03 to 0.13), 0.02 at ten (−0.01 to 0.04), and 0.02 at twenty (0.01 to 0.04). Against these values, pool-basis whitening exceeds the oracle arm by 2.58 points at five patients (2.25 to 2.89), 2.00 at ten, and 1.26 at twenty. The gap of 2.66 points between cohort-basis and pool-basis whitening at five patients therefore splits

into a spectrum part of 0.028 (0.013 to 0.050) and a direction part of the remaining 97 percent (Figure 2). On BRACS the same decomposition is uninformative in absolute terms, because cohort whitening gains nothing there, but the ordering is the same: the oracle arm stays within 0.24 points of cohort whitening at every patient count, while pool-basis whitening exceeds it by 2.96 points at five patients (1.48 to 4.58).

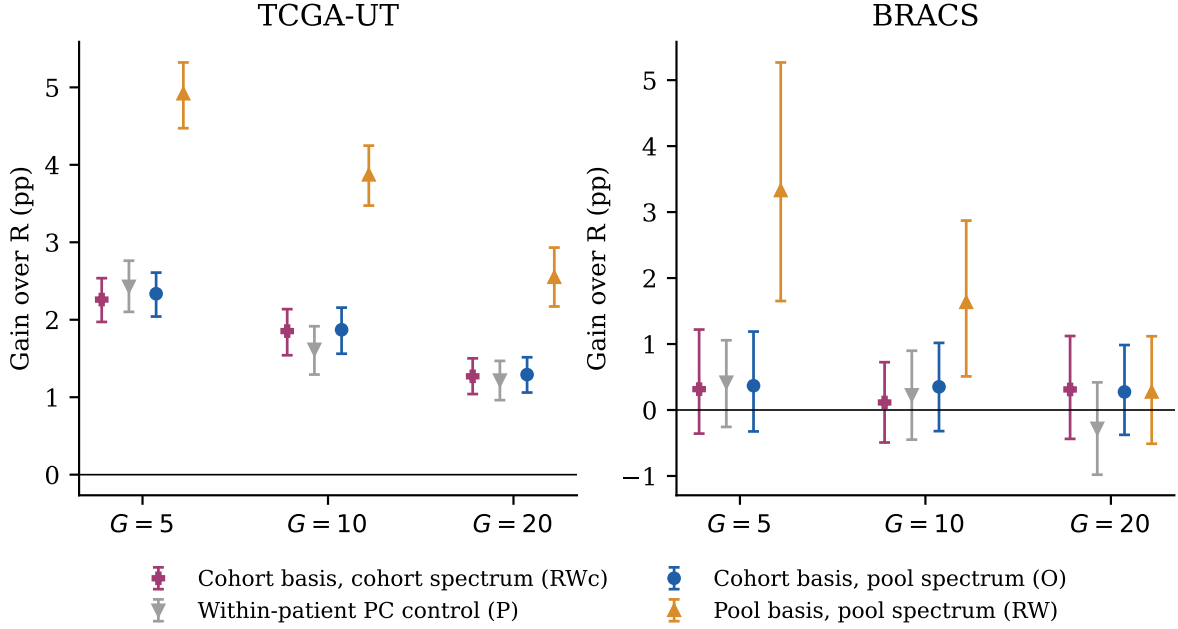


Figure 2: Replacing the cohort’s eigenvalues by the pool variances along the same directions moves accuracy by less than a tenth of a point; replacing the directions moves it by two to three points. Paired gains over the real cohort in percentage points at each patient count. Error bars show the paired 95% bootstrap intervals; markers are offset horizontally within each patient count.

The direction control shows that the between-patient directions are not specific. Whitening along the top within-patient principal components of matched rank, with the cohort eigenvalues attached in rank order, matches cohort whitening on both datasets: the contrast $RWc - P$ is -0.17 points at five TCGA-UT patients (-0.44 to 0.10), 0.24 at ten, and 0.05 at twenty, and on BRACS it ranges from -0.12 to 0.59 points; every interval includes zero. At five TCGA-UT patients the control is in fact the numerically best cohort-only arm in the study, at 64.91 percent against 64.74 for cohort whitening, though the difference is well inside its interval. The two subspaces are not unrelated: their overlap is 0.447 at five patients, ten times the 0.047 expected for a random subspace of rank 120 in 2,560 dimensions (Table 3), so the control shows that a second high-variance cohort subspace whitens equally well, not that an arbitrary subspace would.

5.3 Diagnostics and validation selection

The cohort spectrum is over-dispersed in the direction the motivation predicted. The oracle exponent, the slope of $\log v_j$ on $\log \lambda_j$ within a fit, averages 0.816 at five TCGA-UT patients and falls to 0.743 at twenty; on BRACS it averages 0.830 at five patients and approaches one at twenty (Table 3). Every value is below one, so the pool variances along the cohort’s directions are indeed less spread than the cohort’s own eigenvalues, and no value exceeds one, so the grid’s exclusion of $\beta > 1$ is not what produced the null result. The cohort subspace also misses a substantial part of the population structure: it contains 69.8 percent of the pool’s between-

patient variance at five TCGA-UT patients and 57.7 percent on BRACS, rising to 93.8 and 87.6 percent at twenty patients.

	TCGA-UT			BRACS		
	$G = 5$	$G = 10$	$G = 20$	$G = 5$	$G = 10$	$G = 20$
Rank $r = C(G - 1)$	120	270	570	28	63	133
Oracle exponent	0.816	0.783	0.743	0.830	0.909	0.973
Pool variance in cohort span	0.698	0.845	0.938	0.577	0.760	0.876
Subspace overlap with control	0.447	0.544	0.641	0.394	0.454	0.491
Overlap of a random subspace	0.047	0.105	0.223	0.011	0.025	0.052
Selected $\beta = 0$	0	0	0	8	10	4
Selected $\beta = 0.25$	0	0	0	5	4	3
Selected $\beta = 0.5$	0	0	0	5	4	7
Selected $\beta = 0.75$	1	2	12	7	2	5
Selected $\beta = 1$	29	28	18	5	10	11

Table 3: The cohort spectrum is over-dispersed, yet validation almost always keeps it on TCGA-UT. The oracle exponent is the within-fit slope of the log pool variance on the log cohort eigenvalue; the pool-variance share and the subspace overlap are defined in Section 2; the random-subspace row is the reference value $r/2,560$ for the overlap. The first four rows are means over the 30 fits per patient count; the selection rows are counts of those 30 fits.

Validation selection separates the two datasets sharply. On TCGA-UT the tuned arm keeps $\beta = 1$ in 29 of 30 fits at five patients and 28 of 30 at ten, so the selection rule almost never departs from cohort whitening, and where it does the departure does not pay: at twenty patients validation prefers $\beta = 0.75$ in 12 of 30 fits, and the resulting test contrast is -0.02 points, consistent with selection noise rather than with a real benefit. On BRACS the selections spread over the whole grid — 8, 5, 5, 7, and 5 fits across the five exponents at five patients — which is what an uninformative validation signal on 24 validation patients produces, and the corresponding test contrast is -0.11 points.

The whitening scale absorbs part of the compression without repairing it. On TCGA-UT the flat arm shifts its selected scale factor upward, to 10 in 27 of 30 fits at five patients and to 1 or 10 in 27 to 30 of 30 fits at the larger patient counts, whereas the arms at $\beta \geq 0.5$ and the oracle arm select factor 1 in 19 to 24 of 30 fits at five patients and in all 30 at ten and twenty, matching the behaviour of cohort whitening (Queisler, 2026e). Flattening the spectrum thus calls for a stronger contraction of the subspace, and the tuning grid supplies it, yet the arm still loses 0.92 points. On BRACS the smallest factor 0.1 is selected in 12 to 18 of 30 fits in every family, the same near-inactive whitening reported for that dataset by Queisler (2026a); the largest factor 100 is selected in at most 2 of 30 fits anywhere.

6 Discussion

Of the four readings fixed in Section 4, the results select the second and the fourth. The tuned arm matches cohort whitening to within 0.02 points at every patient count on TCGA-UT and to within 0.15 on BRACS, and the oracle arm, which replaces the cohort’s eigenvalues by the population quantity they estimate, adds at most 0.08 points. The spectrum is therefore not the bottleneck of cohort whitening, and the direction control adds that the specific between-patient directions are not required either.

The null result is not a failure to detect over-dispersion. The diagnostics confirm it: the pool variances along the cohort’s directions grow like the 0.74 to 0.82 power of the cohort’s eigenvalues,

so the sample spectrum is spread more widely than the population one, exactly as the random-matrix argument predicts (Marchenko and Pastur, 1967; Johnstone, 2001). What the results show is that this error is inconsequential for the task. Two properties of Equation 2 explain why. The transform depends on the spectrum only through the saturating factors $(1 + \kappa \nu_j)^{-1/2}$, and the scale-free normalization of κ removes any common factor, so a moderate distortion of the eigenvalue ratios changes the transform far less than it changes the covariance estimate. And the arm that gets those ratios exactly right gains 0.08 points, which caps what any rule in this family could deliver, whether a power law, linear shrinkage (Ledoit and Wolf, 2004), or the direction-wise corrections of nonlinear shrinkage (Ledoit and Wolf, 2020). Better covariance estimation is a solved problem that this application does not have.

What the cohort lacks is directions. Its rank-120 subspace at five patients contains 70 percent of the pool’s between-patient variance, and closing the remaining gap to the pool basis is worth 2.58 points, thirty times the entire spectrum headroom. The same asymmetry holds at ten and twenty patients, and on BRACS, where the cohort subspace holds only 58 percent of the pool variance at five patients and cohort whitening gains nothing at all. This sharpens the conclusion of the preceding studies. Cohort whitening, directional shrinkage (Queisler, 2026d), and centre-uncertainty training (Queisler, 2026c) converge at about one to two points not because they estimate the same covariance imprecisely, but because they can only act inside the span that $C(G - 1)$ patient deviations reach. A remedy that improves on them must widen that span, which requires between-patient structure from outside the labelled cohort — unlabelled slides, related cohorts, or a population covariance carried by the feature extractor — rather than a better-conditioned estimate from inside it.

The direction control refines the mechanism statement of the earlier reports in a way that was not anticipated. Whitening along the top within-patient principal components performs as well as whitening along the between-patient eigenvectors, at matched rank and with the same eigenvalues, on both datasets. The two subspaces overlap by 0.39 to 0.64, far above chance, so this is not evidence that any subspace works; the high-variance directions of patch-level variation within patients and the directions along which patient means differ are partly the same directions. It does mean that the effective description of the remedy is a contraction of a high-variance low-rank subspace estimated from the cohort, and that the between-patient construction, which is the more expensive of the two and needs the patient labels, is not what makes it work.

7 Limitations

The oracle arm is an oracle for one target, not a maximum over all spectra. It sets each eigenvalue to the pool’s between-patient variance along that direction, which is the quantity the cohort eigenvalue estimates, but a spectrum chosen to maximize test accuracy on the same basis could exceed it. Such a search was not run, and the claim that the spectrum contributes 0.08 points is therefore a claim about estimating the intended quantity correctly, not about the best conceivable reweighting of the cohort’s directions.

The power family is one-parameter and monotone, so it cannot represent a correction that moves small and large eigenvalues in opposite directions, as nonlinear shrinkage does. That family was not fitted; the argument against it rests on the oracle bound, which applies to any fixed-basis spectrum aimed at the pool variances. The grid stops at $\beta = 1$, and exponents above one were not tested. The oracle exponents, all between 0.74 and 0.97, point away from that region, but this is a diagnostic rather than a test, and the diagnostic summarizes a scattered relation by a single slope.

The direction control is matched in rank and in eigenvalues but overlaps the cohort basis substantially, so it separates the between-patient construction from high-variance directions in general, and not from arbitrary directions. A perturbation matched per direction, for example

a random subspace drawn inside the span of the pool covariance, was not run. All arms reuse the three evaluation splits of the source studies, so they share test patients with every earlier comparison in this series, and repeated use of these partitions inflates the chance that some arm appears to work by selection alone. On BRACS the host remedy gains nothing, so the spectrum contrasts there test a modification of an inactive treatment, and the intervals, which span roughly one point for the power arms and three for the direction contrast, admit effects that the design cannot resolve. The results are conditional on frozen Virchow2 features, multinomial logistic regression, 160 patches per class, and the two cohorts studied.

References

- Iain M. Johnstone. On the distribution of the largest eigenvalue in principal components analysis. *The Annals of Statistics*, 29(2):295–327, 2001. doi: 10.1214/aos/1009210544.
- Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004. doi: 10.1016/S0047-259X(03)00096-4.
- Olivier Ledoit and Michael Wolf. Analytical nonlinear shrinkage of large-dimensional covariance matrices. *The Annals of Statistics*, 48(5):3043–3065, 2020. doi: 10.1214/19-AOS1921.
- Vladimir A. Marchenko and Leonid A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967. doi: 10.1070/SM1967v001n04ABEH001994.
- Yannik Queisler. Patient shortage in histopathology: Centre error carries the patient benefit on bracs, but the whitening channel does not separate. Technical report, TU Berlin, 2026a. Companion BRACS mechanism report, Experiment 18.
- Yannik Queisler. Patient shortage in histopathology: Class-centre error explains most of the benefit of additional patients. Technical report, TU Berlin, 2026b. Companion centre-error report, Experiment 16.
- Yannik Queisler. Patient shortage in histopathology: Training against class-centre uncertainty gains 2.3 points on tcga-ut and nothing measurable on bracs. Technical report, TU Berlin, 2026c. Centre-uncertainty report, Experiment 21.
- Yannik Queisler. Patient shortage in histopathology: Directional shrinkage of class centres gains about one point on tcga-ut and nothing measurable on bracs. Technical report, TU Berlin, 2026d. Directional-shrinkage report, Experiment 20.
- Yannik Queisler. Patient shortage in histopathology: Whitening along between-patient directions is a separate, additive channel. Technical report, TU Berlin, 2026e. Companion patient-directions report, Experiment 17.
- Donald B. Rubin. The bayesian bootstrap. *The Annals of Statistics*, 9(1):130–134, 1981. doi: 10.1214/aos/1176345338.
- Eric Zimmermann, Eugene Vorontsov, Julian Viret, Adam Casson, Michal Zelechowski, George Shaikovski, Neil Tenenholtz, James Hall, Thomas Fuchs, Nicolo Fusi, Siqu Liu, and Kristen Severson. Virchow2: Scaling self-supervised mixed magnification models in pathology. *arXiv preprint arXiv:2408.00738*, 2024. doi: 10.48550/arXiv.2408.00738.